

Collins

M

# Student's Book 3



# Contents

## Topic 1 Living and non-living things 1

- 1.1
- 1.2
- 1.3
- 1.4
- 1.5

Looking back Topic 1

## Topic 2 Plants 13

- 2.1
- 2.2
- 2.3
- 2.4
- 2.5
- 2.6

Looking back Topic 2

## Topic 3 Humans and other animals 27

- 3.1
- 3.2
- 3.3
- 3.4
- 3.5
- 3.6
- 3.7
- 3.8

Looking back Topic 3

## Topic 4 Materials 45

- 4.1
- 4.2

Looking back Topic 4



5

### Looking back Topic 5

65

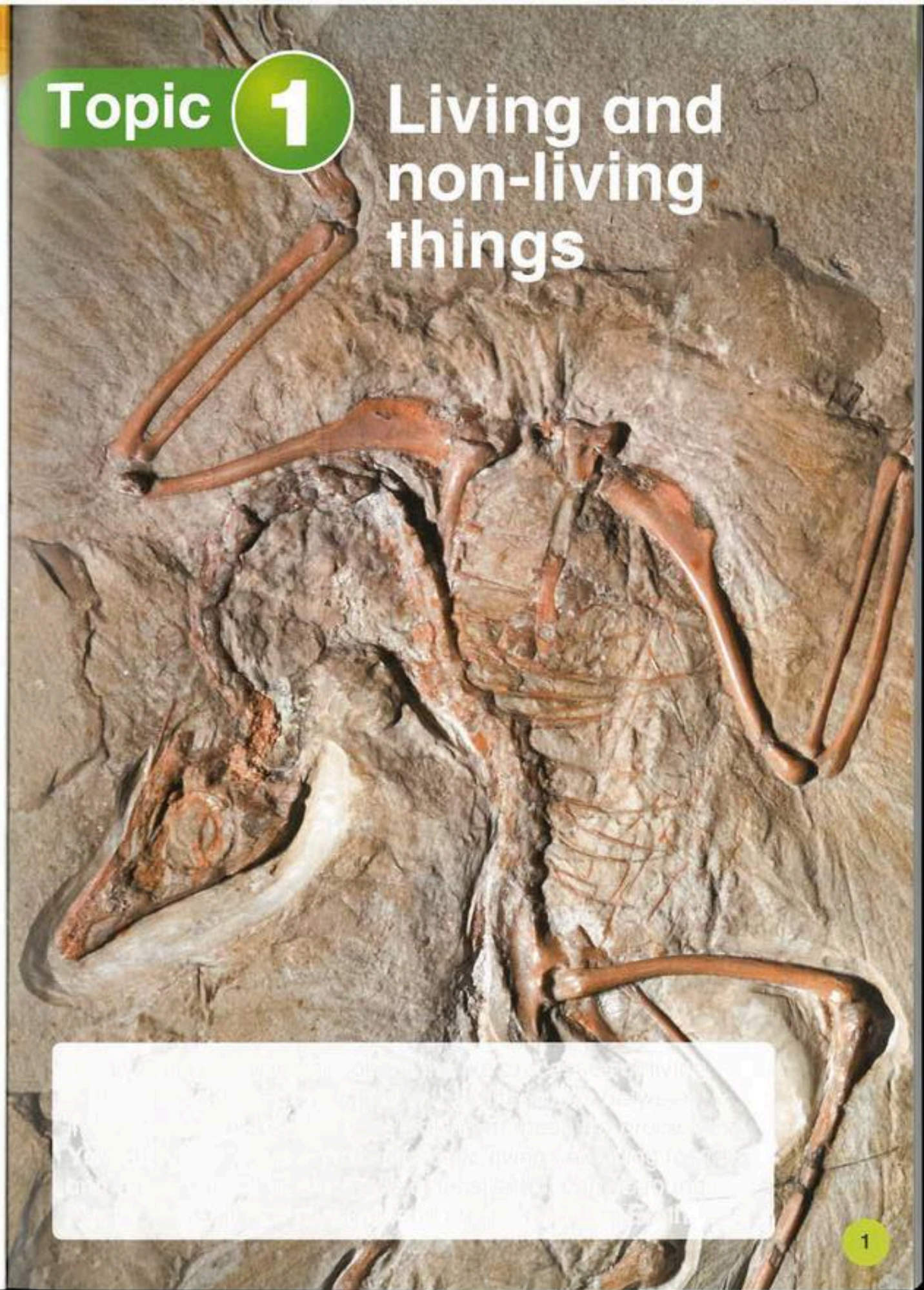
### Glossary

97

## Topic

1

## Living and non-living things





# 1 ry

• • • •

1

2



3

| Topic | Question | Answer | Notes |
|-------|----------|--------|-------|
|       |          |        |       |
|       |          |        |       |
|       |          |        |       |

2

## y words

- 
- 
- 
- 
- 
- 
- 
- 

T 1

4

5



6

7



1

2

3

1

-



L

1



2

3

4

# Key words

- long
- short
- wide
- narrow

1

2

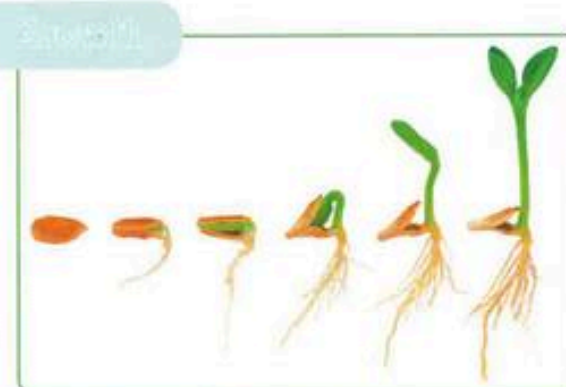
Movement



Reproduction



Growth



Roundness



1

2

3

- Animals and plants are taking things.
- Life processes include movement, growth, roundness and reproduction.

5



# 1.3 Is iv g

1



2



3

4



2

1

| Living things | Non-living things | Things that are both |
|---------------|-------------------|----------------------|
|               |                   |                      |

3

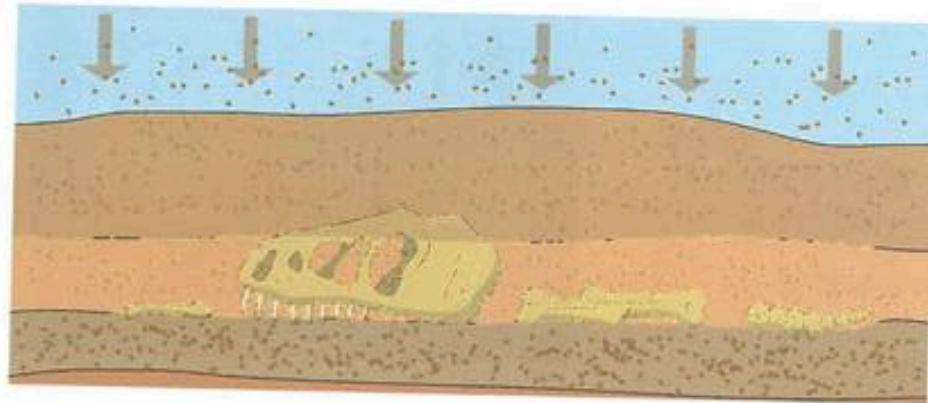
- Some things are living, and some are non-living.
- Plants and animals are living things.
- Houses and cars are non-living things.

iving things

1



## 4 Fossils



### Key words

- fossil
- fossil fuel
- fossil
- fossil
- fossil
- paleontologist



1

2

1

3

A B

4

5



1

2

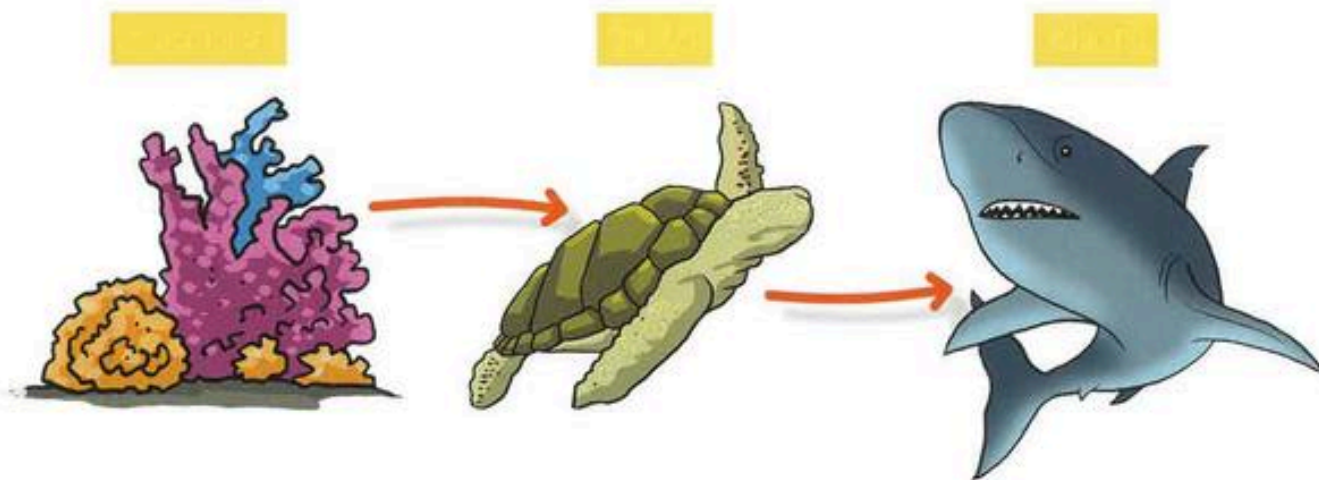
3

- Fossils can be found in many different places, including in the ground, in rocks, and in the sea.
- Fossils can be used to learn about the history of the Earth and the life that has lived on it.
- Fossils can also be used to study the changes in the environment over time.



# 1.5 F

corn → chicken → fox



1  
2



## y words

- feeding
- relation
- predator
- producer
- nutrition
- omnivore
- organism
- ecological

1

Living

4



5

1

Blank box for notes.

2

Blank box for notes.

3

Blank box for notes.

- Food chains show the feeding relationships in an ecosystem.
- All food chains start with a producer.
- Animals are consumers because they eat plants and other animals.



## Looking back Topic 1

I remember you

- Animals and plants are living things.
- They need water, food, shelter, space to live, movement and growth.
- We can protect them by taking care of the environment.
- Some animals and plants are endangered. Some non-living things have never lived.
- Plants are the food for many animals and birds.
- Plants change with the seasons and the weather.
- Some plants are the same as the ones we see in the garden.
- All living things are part of the environment.

## How well do you remember?

1



2

3

4

5



## Topic 2 Plants

In this topic you will learn about the different parts of plants. You will learn how water and light to grow. You will learn how water and light to grow. Finally, you will learn how water and light to grow.



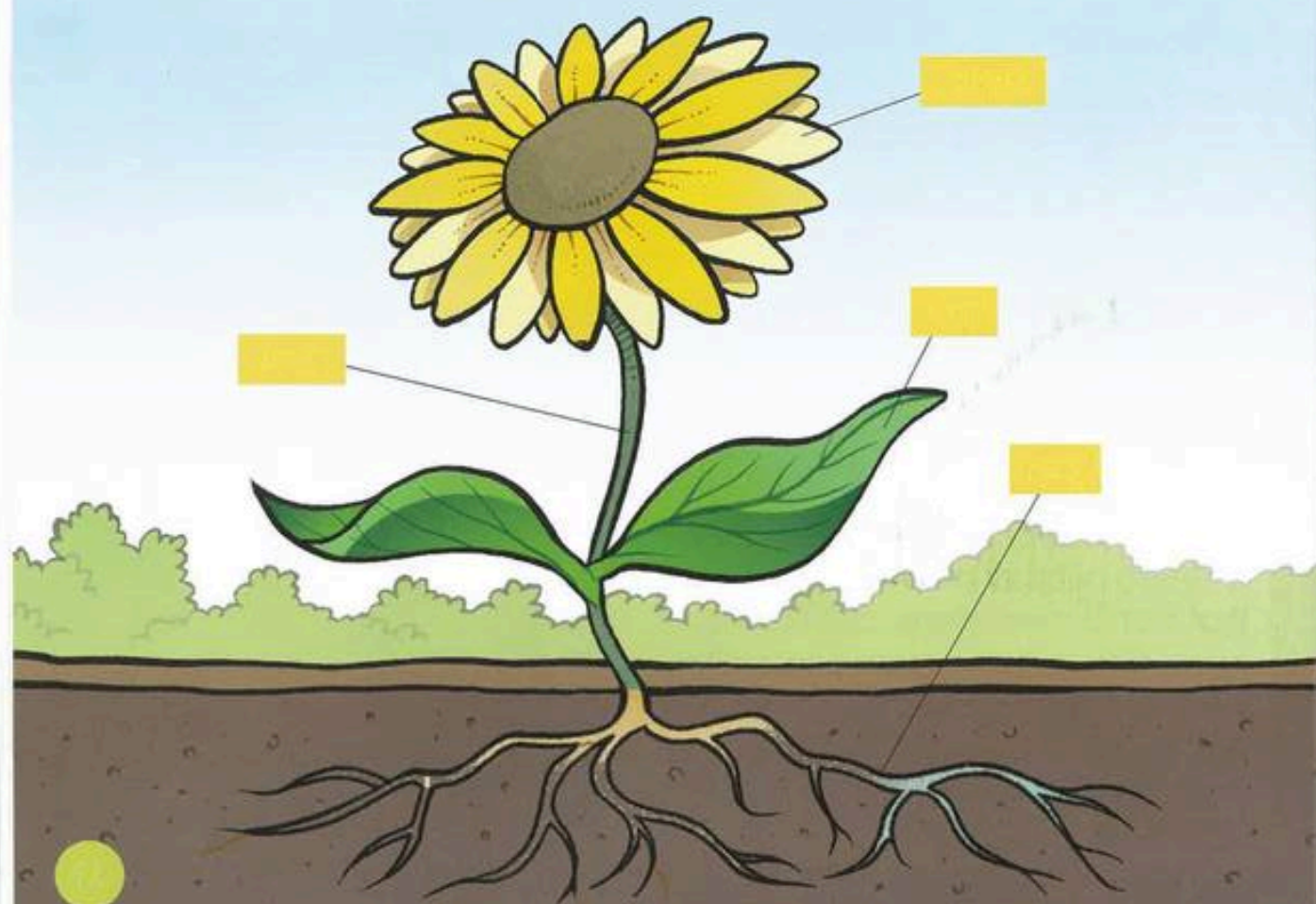
1 P

a

## Key words

- roots
- stems
- leaves
- flowers

- 1 Where does the plant take water from?
- 2 Trace the path of water into the leaves. ▼
- 3 What is the main part of the plant?



1

A

B



A

B



2

3

- The roots are the main part of the plant.
- They are in the soil and take water from it.
- The water goes up the stem to the leaves.



## 2.2 Plant with

### Key words

- 
- 
- 



A

1. Name the plant. A B

2. What kind of soil does it grow in?

3. What kind of water does it need? Think each plant has? Explain your answer.

B



1

2

|                                                                                | Plant 1 | Plant 2 | Plant 3 |
|--------------------------------------------------------------------------------|---------|---------|---------|
| 1. Name the plant.                                                             |         |         |         |
| 2. What kind of soil does it grow in?                                          |         |         |         |
| 3. What kind of water does it need? Think each plant has? Explain your answer. |         |         |         |



3

- Plants need water to grow.
- Water is taken in through the roots and transported through the xylem to the leaves.

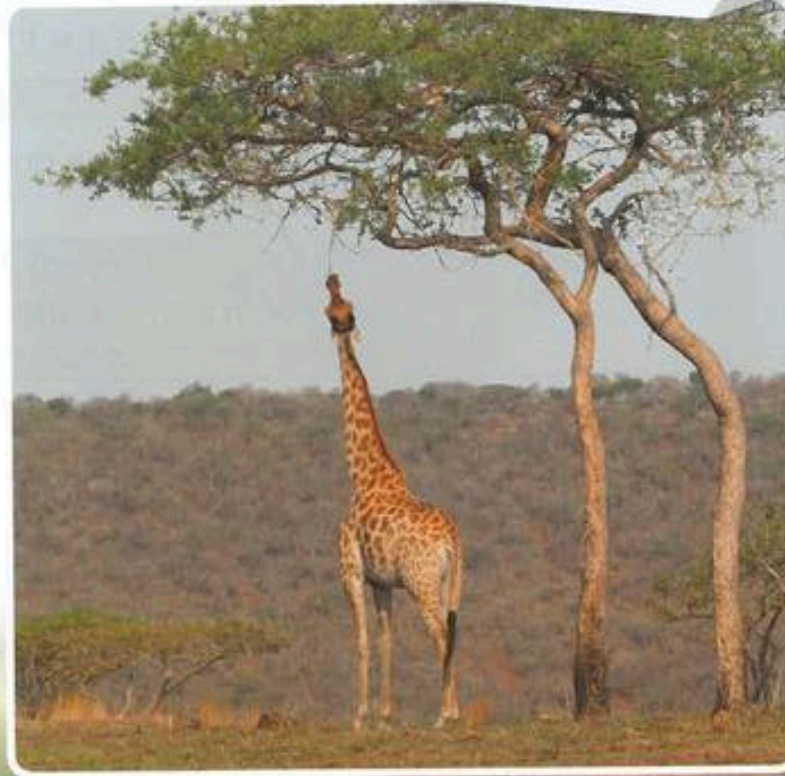


## 2.3 Pla

g

key words

- ...



2

1

2

3



1

2

3

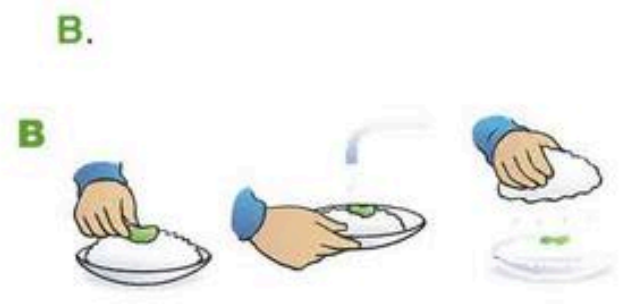
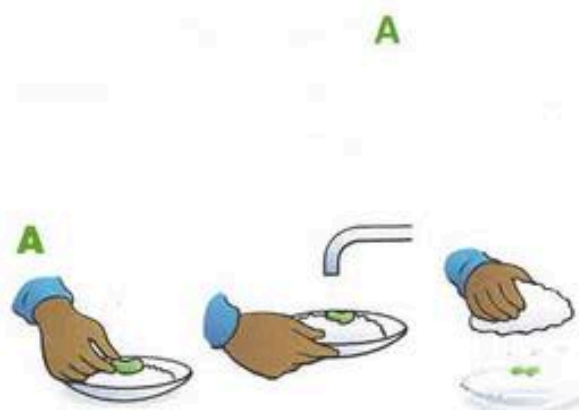
- ...
- ...



## 2.4 P a w

### Key words

- ...
- ...



- 1
- 2
- 3
- 4

B

2

and the seeds had been waiting for 10,000 years for the water they needed.



1

2

3

- Plants grow in order to live.
- Plants need water to grow.



## 2.5 Healthy



1

2

3



without  
fertiliser



4

5

6

- Plants grow healthy and green, and produce good yield.
- Weeds are removed and soil is rich.
- We can produce 1 ton of corn per hectare.

1

2

3



## 2.6 flow g

### Key words

- nest
- pollen
- honey
- bee
- hive



1

2

- A. The bee is very busy and it is very busy.
- B. The bee is very busy and it is very busy.
- C. The bee is very busy and it is very busy.
- D. The bee is very busy and it is very busy.
- E. The bee is very busy and it is very busy.

3



4

1

2

3

- The bee is very busy and it is very busy.
- The bee is very busy and it is very busy.
- The bee is very busy and it is very busy.



## Looking back T 2

you

- Plants reproduce and grow.
- Each part of the plant has a different job.
- The main part of the plant is the root, which takes in water and food.
- The stem holds the plant up and carries water and food to the leaves.
- The leaves make food for the plant using sunlight and water.
- Water is transported up the stem from the roots to the leaves.
- Food is transported down the stem from the leaves to the roots.
- Plants and other animals need plants for food.
- Plants need healthy roots, leaves and stems to grow well.
- Some seeds travel from the leaves of a plant.
- Some animals eat the leaves of a plant.
- The way a plant grows is affected by its environment.
- Some plants grow in water, some in soil, some in rocks.

### How well do you remember?

- 1
- 2
- 3
- 4
- 5

## Topic

3

## Humans and other animals



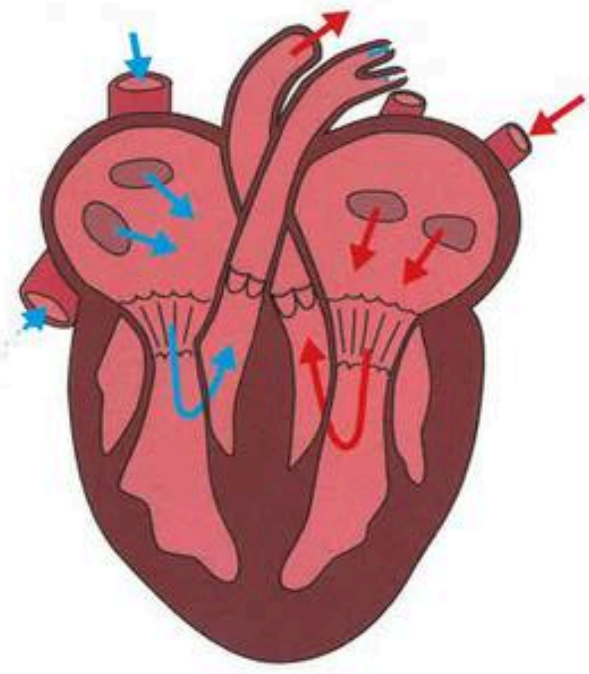
Humans and other animals need plants for food. Plants make food using sunlight and water. The food is transported down the stem to the roots. Some plants grow in water, some in soil, some in rocks. The way a plant grows is affected by its environment.



# 3.1 M a ntoti

Key words

- 
- 
- 
- 



3

4

1

2

3

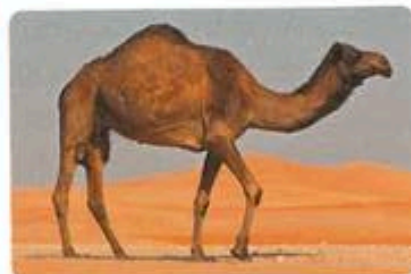
•



## 3.2 Classifying a

### Key words

- shark
- eagle
- camel
- crocodile
- mantis
- frog
- amphibian



1

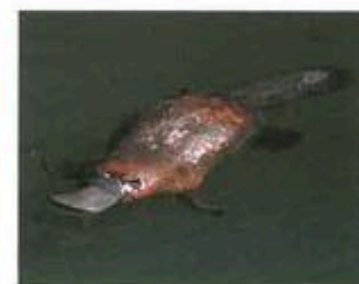
2

3

4

1

3



- Sea turtles are found in all oceans and are endangered.
- Platypus is a unique animal found only in Australia.
- Manatees are found in shallow coastal waters and are also endangered.



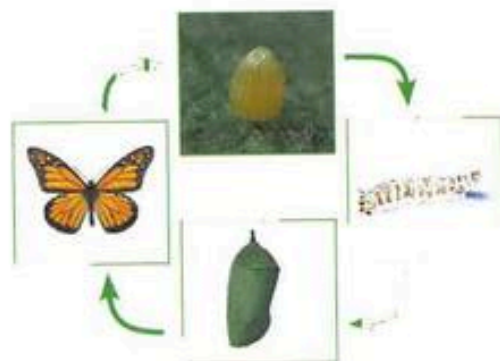
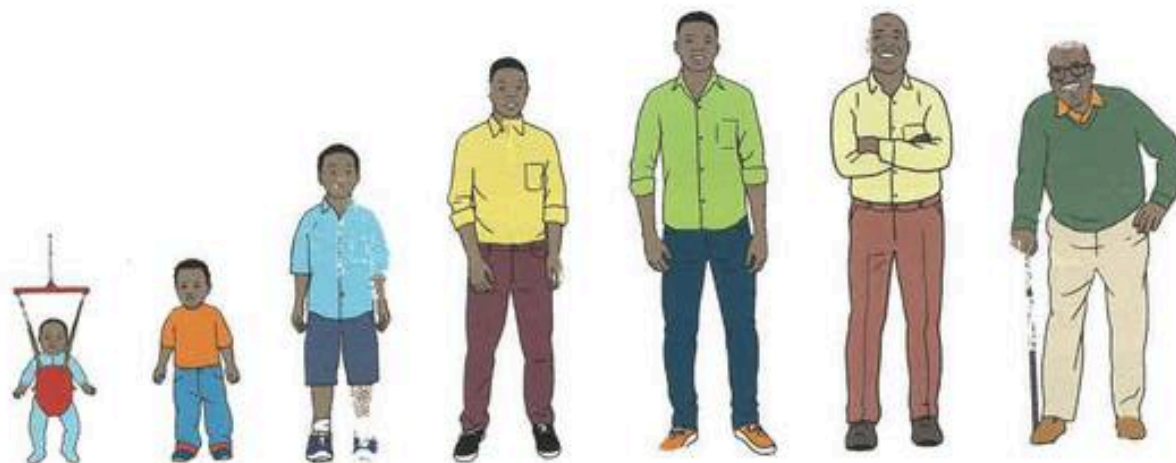
### 3.3 Grow g a cha g g

#### Key word

- *growing*
- *change*

1

2



4



1

2

3

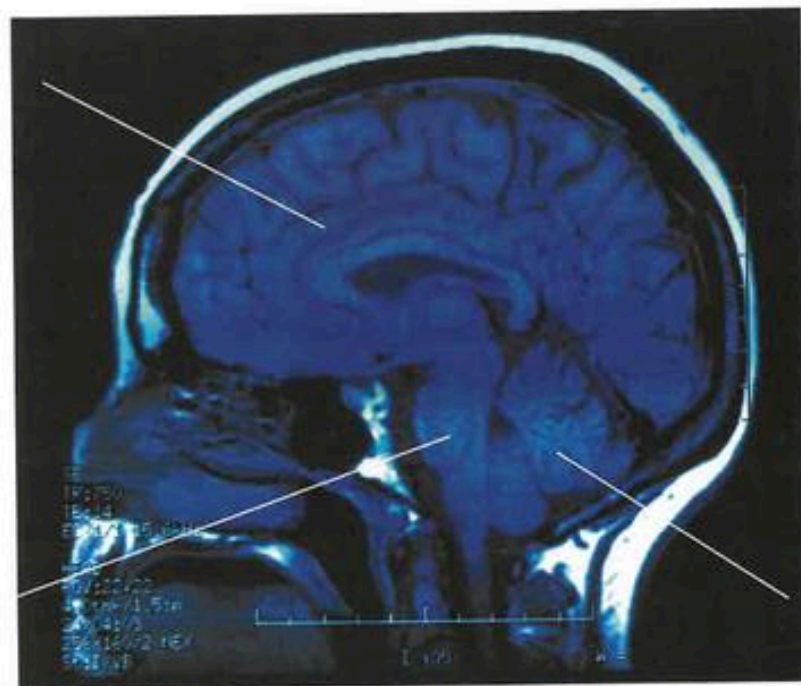
- We can see how things grow and change.
- Things grow and change, but they are always different when they are born.



## 3.4 Y bra

### Key words

- ...
- ...
- ...



1  
2



4

1

2

3

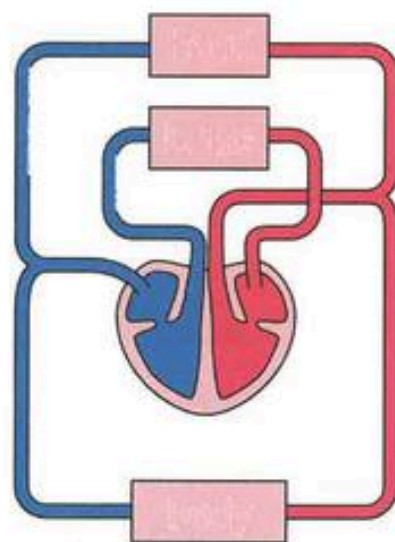
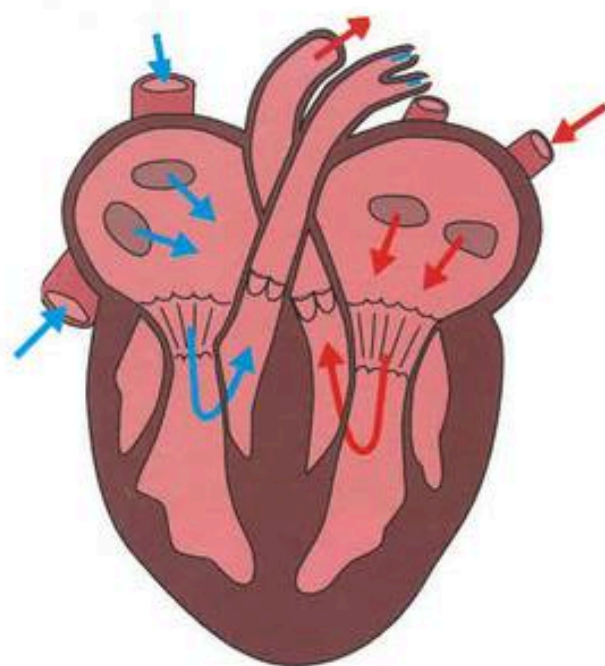
- The ...
- The ...
- The ...



# 3.5 Y

## Key word

- muscles
- blood
- lungs
- pulse rate



- 1
- 2
- 3

4



1

2

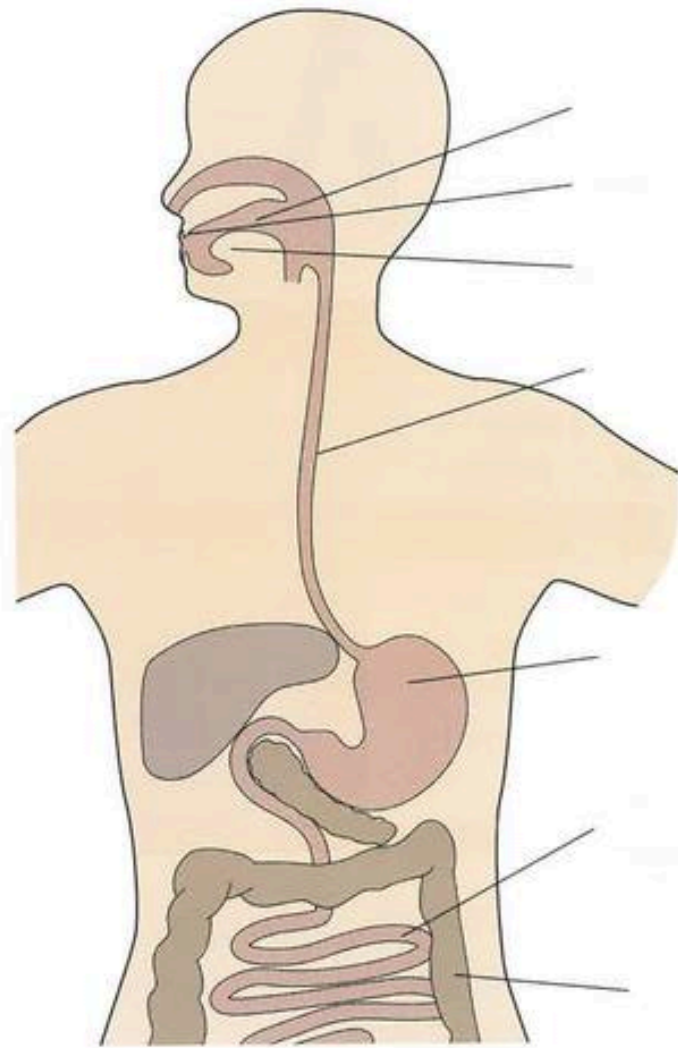
3

- The heart is a pump made of muscle.
- The heart pumps blood around our body.
- The heart pumps blood from the lungs to the rest of the body.



### 3.6 Y a a

1



#### word

- The digestive system starts in the mouth with chewing food.
- Food then goes to the stomach where it is broken down into smaller pieces.
- Next, the food goes to the small intestine.
- Finally, the food travels through the large intestine.

2

3

4



1

2

3

- The digestive system starts in the mouth with chewing food.
- Food then goes to the stomach where it is broken down into smaller pieces.
- Next, the food goes to the small intestine.
- Finally, the food travels through the large intestine.



# 3.7 Y g



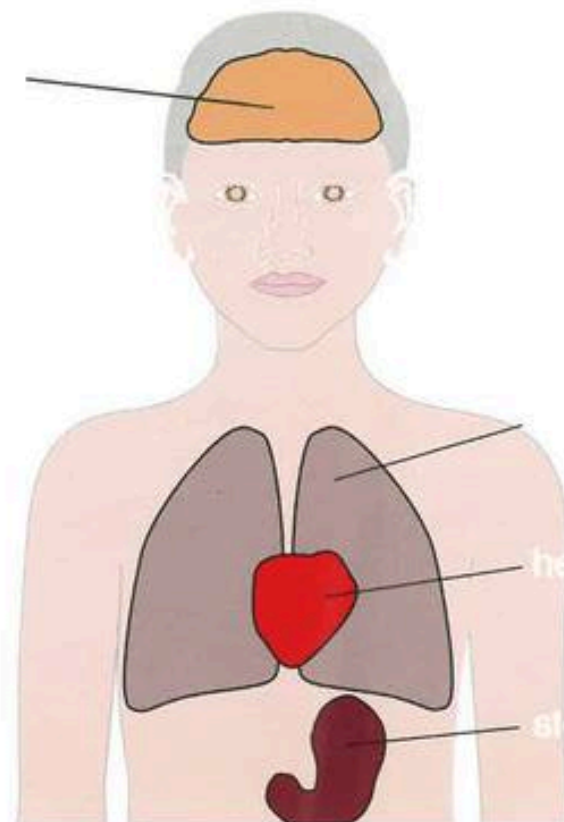
1

2



## word

- lungs
- trachea
- bronchi
- alveoli
- diaphragm
- pleura



3

1

3

- Lungs take oxygen from the air and send it to the rest of the body.
- The oxygen is produced by the mitochondria.



## 3.8 gery

### Key words

- surgery
- operation
- procedure
- treatment
- therapy
- intervention
- medical procedure
- surgical procedure

1



2



3



### learned

- Surgery is a medical procedure that involves the use of instruments to treat a disease or injury.
- It is a common way to treat a variety of conditions, including cancer, heart disease, and injuries.
- Surgery can be performed on a variety of parts of the body, including the brain, heart, and lungs.

1

2

3



## Looking back Topic 3

you

- You have learned about the different types of materials.
- You have learned about the properties of different materials.
- You have learned about the uses of different materials.
- You have learned about the different types of materials.
- You have learned about the properties of different materials.
- You have learned about the uses of different materials.
- You have learned about the different types of materials.
- You have learned about the properties of different materials.
- You have learned about the uses of different materials.
- You have learned about the different types of materials.
- You have learned about the properties of different materials.
- You have learned about the uses of different materials.

## How well do you remember?

1



2

3

4

5

## Topic 4 Materials



In this topic you will discover that materials can be used in many different ways. You will learn about some of the materials that are used in the construction of buildings and bridges. You will also learn about the materials that are used in the production of goods and services.



## 4.1 Typhoid fever

**Key word**

- Salmonella
- Bacteria
- Food
- Water

1



2



3



4



1

2

3

- Salmonella is a bacteria that causes typhoid fever.
- It is spread by food and water that has been contaminated with the bacteria.



## 4.2 Materia ro Ea

### Key word

- ...
- ...
- ...
- ...
- ...



in



1

2

3



1

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |

2

3

•

•



## Looking back T 4

you

- What are the three states of matter?
- How do the particles in each state move?
- How do the particles in each state compare in size and mass?

### How well do you remember?

1

2

3



4

5

6

## Topic 5 States of matter

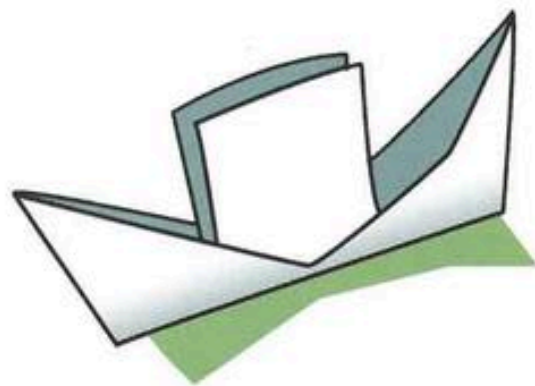
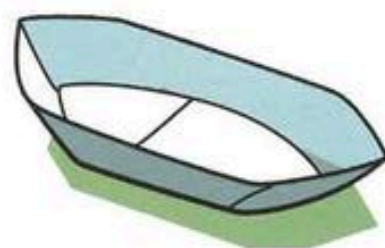
In this topic, you will compare solids and liquids. You will learn about the particles in each state and how they move. You will also learn about the forces between particles. You will learn about the forces between particles in solids and liquids. You will also learn about the forces between particles in gases.



## 5.1 ids

### Key word

- conductivity



1  
2  
3

4



5

6

8

1

3



- The temperature of the water and the surrounding air for making the paper boat float.
- The temperature of the water and the surrounding air for making the paper boat float.



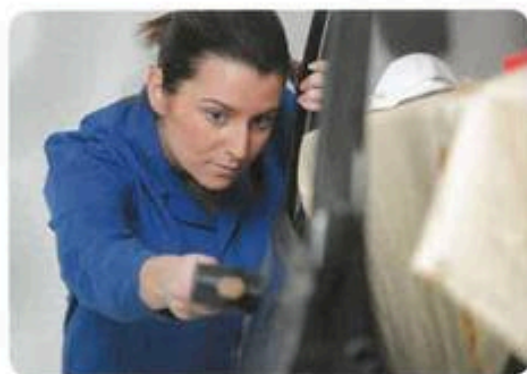
## 5.2



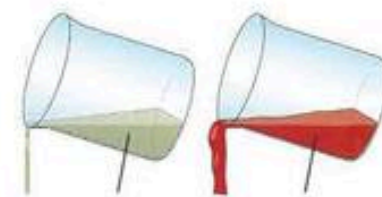
1



2



3



1

|  |  |
|--|--|
|  |  |
|  |  |

2

|  |
|--|
|  |
|  |
|  |

3

|  |
|--|
|  |
|  |
|  |

- Some liquids have a high surface tension.
- Some solids are very hard and strong.
- Some solids are made of many small particles that are held together by strong forces.
- Some solids are made of many small particles that are held together by weak forces.
- Some solids are made of many small particles that are held together by very weak forces.
- Some solids are made of many small particles that are held together by no forces at all.



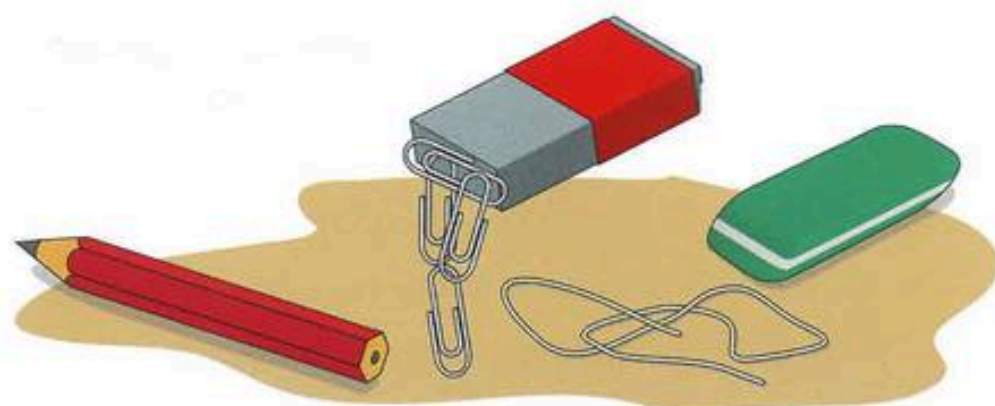
## 5.3 Mix

### Key words

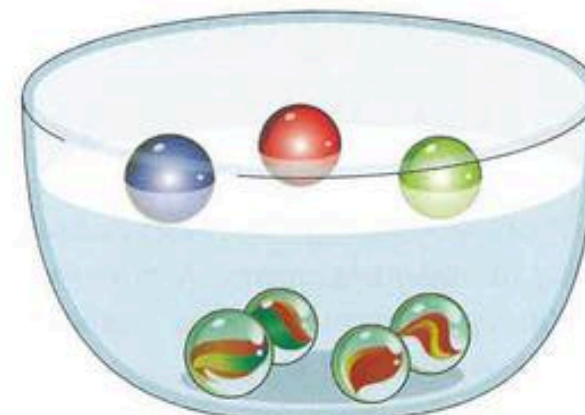
- mixture
- retain
- mixing
- filtering
- magnet



2



3



1



3

- A mixture is a combination of two or more substances that are not chemically combined.
- The substances in a mixture retain their own properties.
- The substances in a mixture can usually be separated.
- Some mixtures can be separated by simple methods like filtering, sieving, or magnetism.



## 5.4 M x i a g

key word



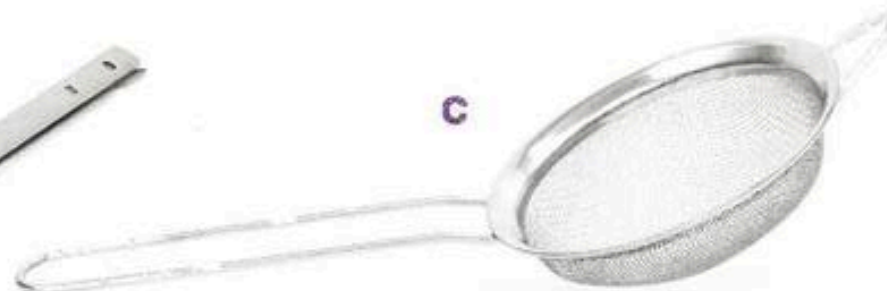
1



A



B



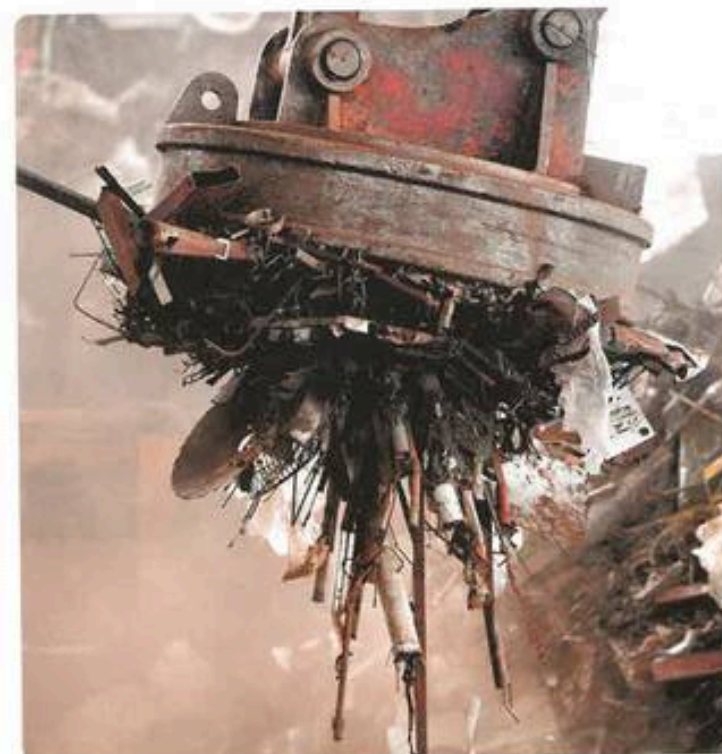
C



D

2

3



4

▲

1

2

3

- The mechanical harrow is a type of agricultural machine used for breaking up soil and removing weeds.
- It is used in the preparation of the soil for sowing seeds.
- A harrow is a type of agricultural machine used for breaking up soil and removing weeds.



5.5

d

qu

# Key words

- ...
- ...



2



3

4

1

3

- ...
- ...



## 5.6 Dissolving

### Key words

- solvent
- solute



1  
2  
3

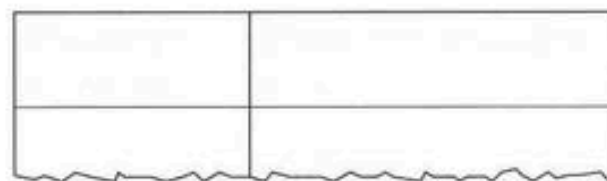
32

5

4

### Activities

1



2

3

### Remember

- A solvent is a liquid that dissolves a solid or a gas.
- A solute is a solid or a gas that is dissolved in a liquid.
- If a substance is dissolved in a liquid, the liquid will be called a solution. If it is a gas, it will be called a gas.

33



## Looking back T

you

- Sketch the shape of a force and label it.
- Explain the shape of a force and the direction they are pointing and what the force would do.
- Draw a force on a model of a force and label it.
- Explain the shape of a force and the direction they are pointing and what the force would do.
- Explain the shape of a force and the direction they are pointing and what the force would do.
- Explain the shape of a force and the direction they are pointing and what the force would do.

## How well do you remember?

1



2



3

4

5

## Topic 6

## Forces, light and magnets





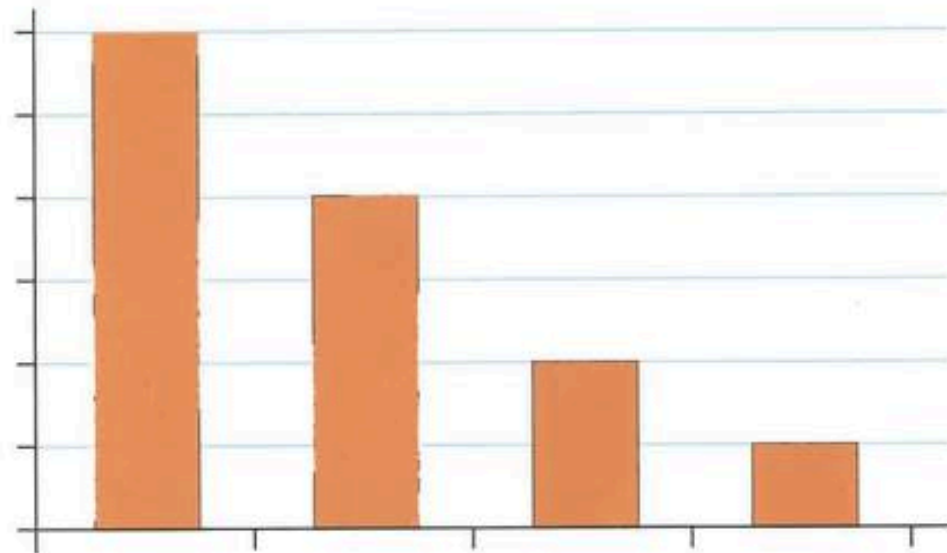
# 6.1 F Investig

## Key words

- ...
- ...



2



4

5

1

2

3



- ...



## 6.2 Gravity – a r a r a force

Why do all things fall down? If you are in a hot air balloon and you jump up into the air, where do you land? Gravity is a natural force which pulls everything towards the centre of the Earth. When you throw a ball up into the air, you can see it comes back down. It's gravity which keeps you on the ground and makes things fall from upwards straight down.



1. What would happen if there was no gravity on Earth?

Without gravity we would fly up into the air and if we wanted to go up, the air would not be able to go up.

2. Why do skydivers use a parachute to slow down their fall?

The increased force of gravity pulls them down to the ground. When the parachute opens, the air resistance slows them down.



### Key words

- gravity
- force
- pull



3

4

1

2

3

- Gravity is a force that pulls everything down towards the centre of the Earth.
- Gravity is a force that pulls everything down towards the centre of the Earth.



## 6. Measuring



### Key words

- Gravity
- Weight
- Mass
- Newton
- Spring scale



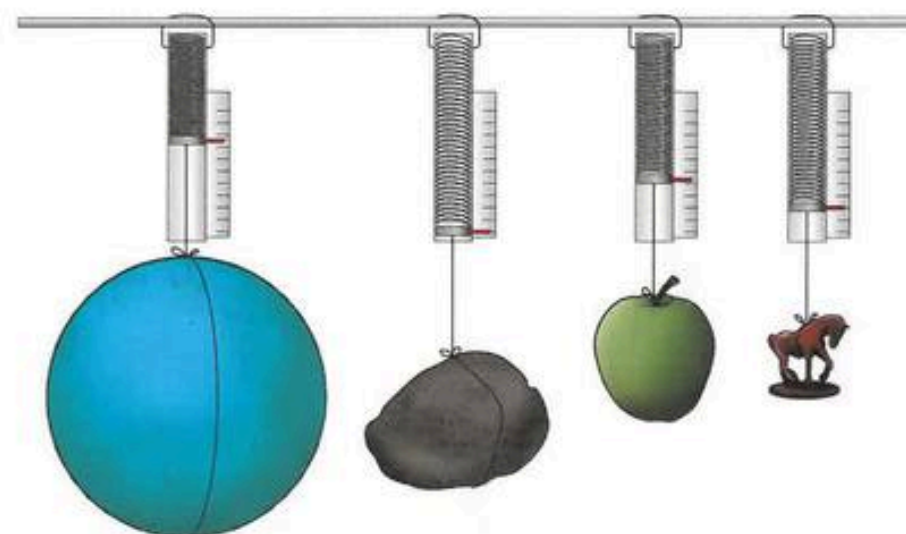
1

2

3

4

6



5

6

1

2

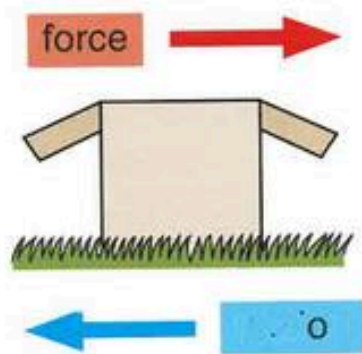
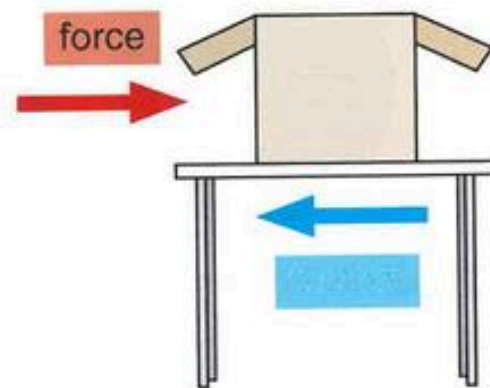
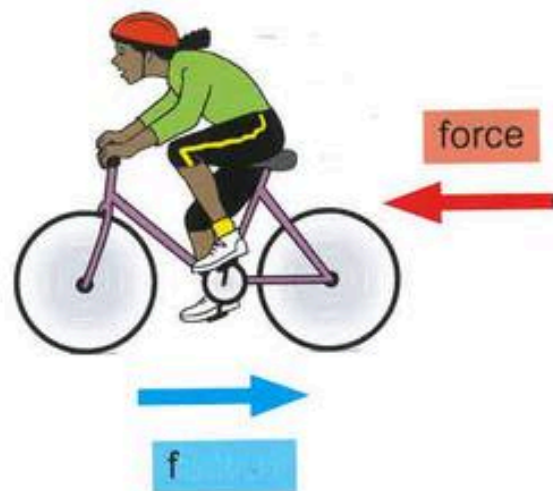
3

- Gravity is the downward force that pulls objects toward the center of the Earth.
- Weight is the force of gravity on an object.
- Mass is the amount of matter in an object.



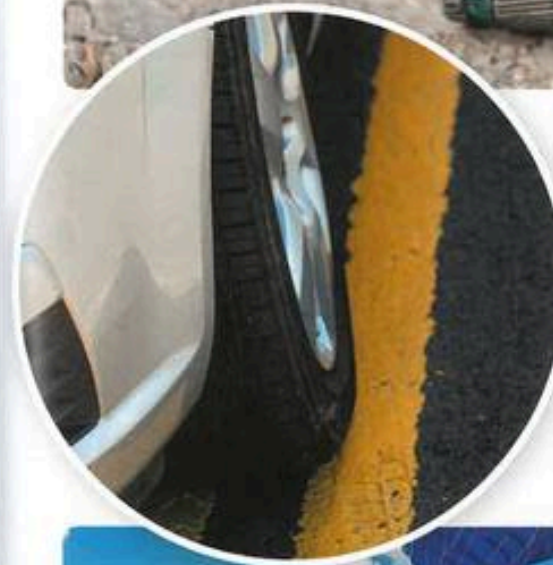
4 F

Key word



2

3



1

2

3

- Friction is a force that acts between two surfaces.
- Friction acts in the opposite direction to the motion.
- Friction is a contact force.



## 6. Will it move?

### Key words

- push
- pull
- sliding



2

3



6

1

2

3

4



- Friction is a force that opposes motion. It is caused by surfaces rubbing against each other.
- Friction is a force that makes it harder to move things. It is caused by dry surfaces and rough surfaces.
- Friction is a force that makes it harder to move things. It is caused by dry surfaces and rough surfaces.
- Friction is a force that makes it harder to move things. It is caused by dry surfaces and rough surfaces.

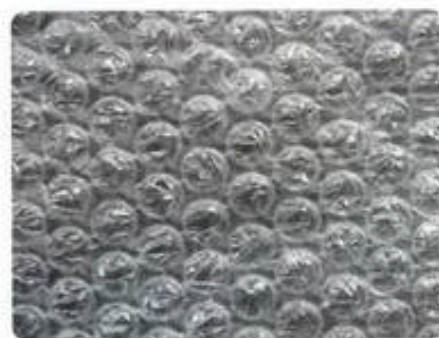
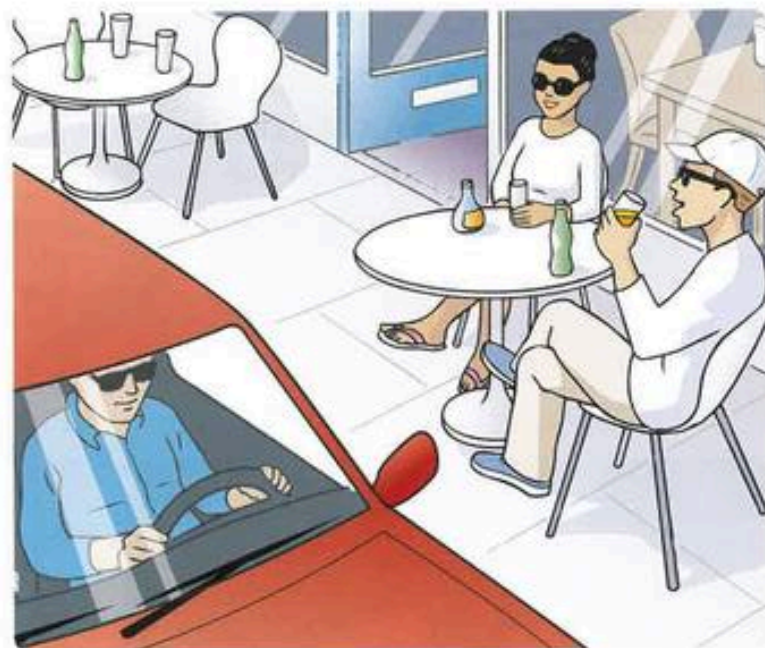


6 L g g

# Key words

- liquid
- gas
- solid
- mixture

2  
3



6

4



3

- Liquid is a substance that has a definite volume but no definite shape.
- Gas is a substance that has no definite volume and no definite shape.
- Solid is a substance that has a definite volume and a definite shape.

77



a w

# Key word

- Light ray
- Shadow



2  
3

6

4



## توضیحات

1

2

3

## نکته

- When the sun is high in the sky, the shadow is straight.
- When the sun is low in the sky, the shadow is spread out.

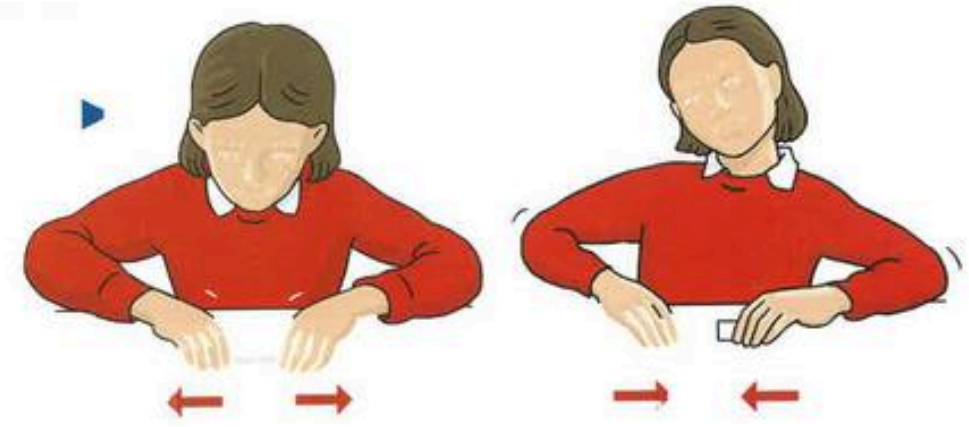
12



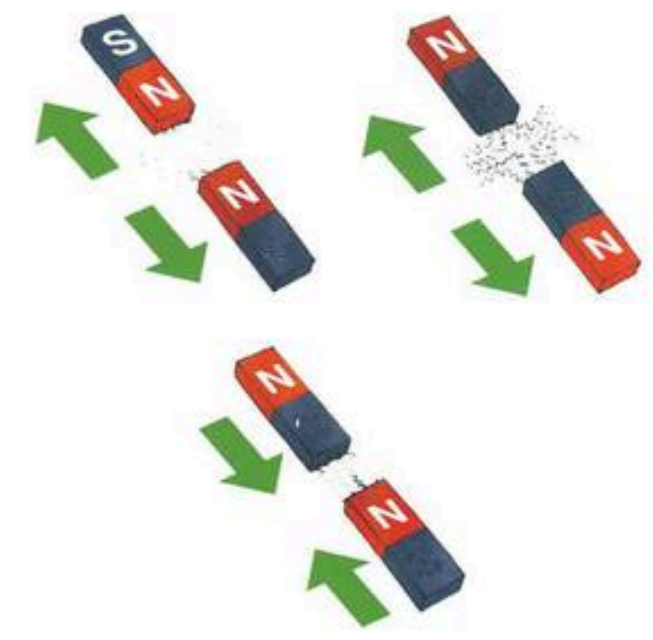
# Mag

**Key words**

- attract
- repel
- pull
- push
- force
- field



2



3

4

1

to

happen

magnets Complete

3

orkbook to

- The force between two magnets is strongest when the poles are close together.
- The force between two magnets is weakest when the poles are far apart.
- The force between two magnets is strongest when the poles are facing each other.

poles).



# Ma a magneti

## Key words

- attract
- repel



1

▲



3

4

1

3

## learned

- Some materials are magnetic.
- Magnets attract certain types of materials.
- Some materials which are not magnetic are called magnetic materials.



## 6.1 The Earth's magnetic field

The Earth's magnetic field is a magnetic field that extends from the Earth's interior to space, where it interacts with the solar wind, a stream of charged particles emanating from the Sun.

1 The Earth's magnetic field is a magnetic field that extends from the Earth's interior to space, where it interacts with the solar wind, a stream of charged particles emanating from the Sun.

The Earth's magnetic field is a magnetic field that extends from the Earth's interior to space, where it interacts with the solar wind, a stream of charged particles emanating from the Sun.

The Earth's magnetic field is a magnetic field that extends from the Earth's interior to space, where it interacts with the solar wind, a stream of charged particles emanating from the Sun.



The  
acted

Earth's magnetic field. From

a map to  
around.



### Key words

- compass
- magnetic field
- true north
- North Pole

3

1



2

a  
rks.



3

- The compass needle is a magnet which is attracted to magnetic north.
- The compass needle always points towards magnetic north.
- True north is 0° and magnetic north is 12.0°.



## Link back Topic 6

## H

2

3

4



## Topic

## 7

# Earth, Moon and Sun

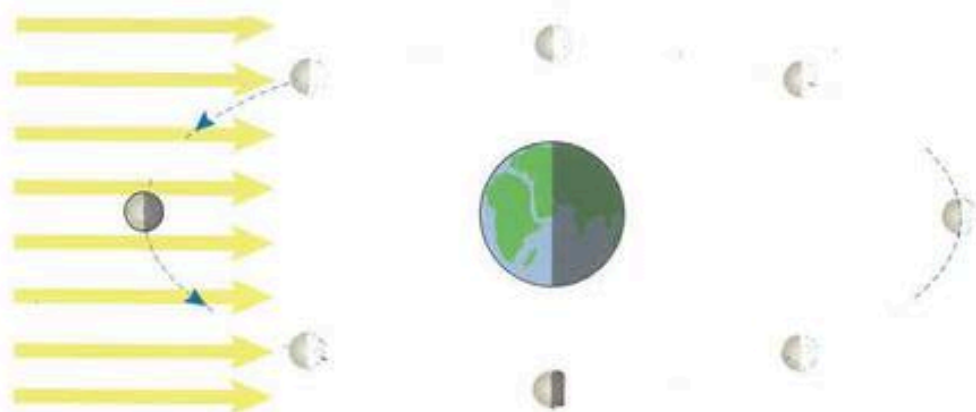




# .1 M cha g

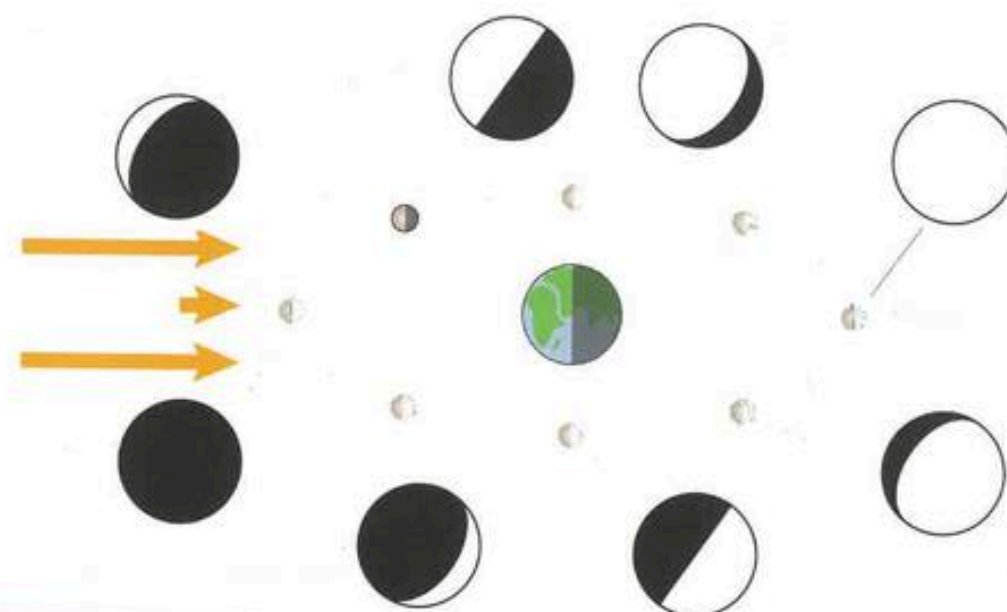
## Key words

- phase
- orbit
- moon



3

4



1

2

3

- The Moon's phases are caused by the Sun's light.
- The Moon's phases are caused by the Moon's position relative to the Earth and the Sun.
- The Moon's phases are caused by the Moon's position relative to the Earth and the Sun.



# Key words

- Sun
- Moon
- Earth
- Orbit
- Revolution
- Rotation

T - ov Ea  
M

3

1

the

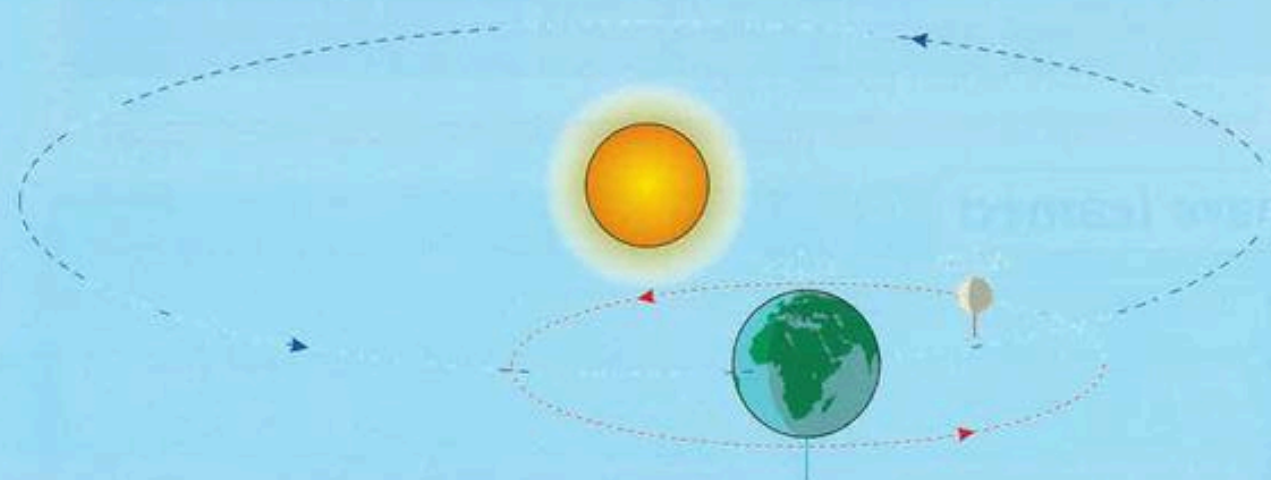
do one

2

3

the  
added  
to show  
nent of the

- The Sun orbits the Earth once every 28 1/2 days
- The Earth orbits the Sun once every 365 1/4 days (one year)
- The Earth rotates once every 24 hours





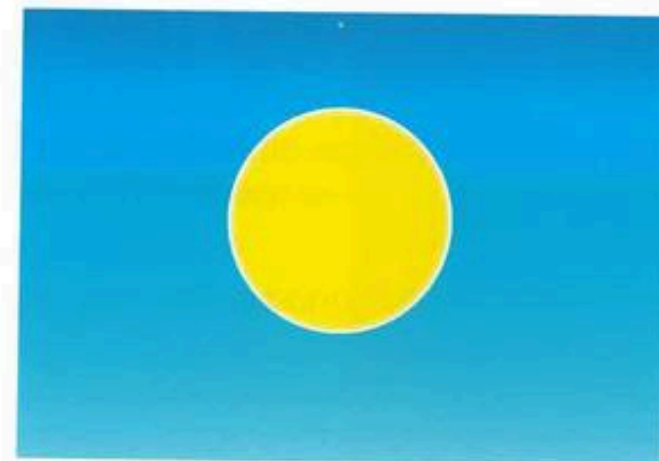
M a Ea  
and

### Key words

- ...
- ...
- ...
- ...
- ...



3



1

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |
|  |  |

3

- ...
- ...
- ...



# 4 A y then a ow

## Key words

- ...
- ...
- ...
- ...
- ...
- ...

1



3



4



## 1. ...

1

2

3

## 2. ...

- ...
- ...
- ...
- ...



## Learning Task 1

you

- The Moon is 384,400 km from the Earth.
- The Moon is 1/4 the size of the Earth.
- The Moon is 1/3 the size of the Earth.
- The Moon is 1/4 the size of the Earth.
- The Moon is 1/3 the size of the Earth.
- The Moon is 1/4 the size of the Earth.
- The Moon is 1/3 the size of the Earth.
- The Moon is 1/4 the size of the Earth.

H

2

4

d  
e

## Glossary





1. The first part of the report discusses the importance of maintaining accurate records of all transactions. This is essential for ensuring the integrity of the financial data and for providing a clear audit trail. The second part of the report details the various methods used to collect and analyze the data, including interviews, surveys, and archival research. The third part of the report presents the findings of the study, which show that there is a significant correlation between the variables being studied. The final part of the report discusses the implications of these findings for future research and for policy-making.

2. The second part of the report details the various methods used to collect and analyze the data, including interviews, surveys, and archival research. The third part of the report presents the findings of the study, which show that there is a significant correlation between the variables being studied. The final part of the report discusses the implications of these findings for future research and for policy-making.

3. The third part of the report presents the findings of the study, which show that there is a significant correlation between the variables being studied. The final part of the report discusses the implications of these findings for future research and for policy-making.

4. The final part of the report discusses the implications of these findings for future research and for policy-making.



5. The first part of the report discusses the importance of maintaining accurate records of all transactions. This is essential for ensuring the integrity of the financial data and for providing a clear audit trail. The second part of the report details the various methods used to collect and analyze the data, including interviews, surveys, and archival research. The third part of the report presents the findings of the study, which show that there is a significant correlation between the variables being studied. The final part of the report discusses the implications of these findings for future research and for policy-making.

6. The second part of the report details the various methods used to collect and analyze the data, including interviews, surveys, and archival research. The third part of the report presents the findings of the study, which show that there is a significant correlation between the variables being studied. The final part of the report discusses the implications of these findings for future research and for policy-making.

7. The third part of the report presents the findings of the study, which show that there is a significant correlation between the variables being studied. The final part of the report discusses the implications of these findings for future research and for policy-making.

8. The final part of the report discusses the implications of these findings for future research and for policy-making.

